

Prathamesh Shete

AI / ML Engineer | Generative AI & Transformer Specialist

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Professional Summary

AI/ML Engineer with **2.6+ years of hands-on experience** in designing, developing, and deploying **production-grade Generative AI and Transformer-based systems**. Strong expertise in **LLMs, RAG architectures, model fine-tuning, and scalable ML pipelines**. Proficient in Python, Hugging Face, TensorFlow, and LangChain, with solid exposure to **AWS deployments, FastAPI microservices, and MLOps practices**. Passionate about building reliable, real-world AI solutions that deliver measurable business value.

Core Skills

Generative AI & Large Language Models

- LLMs, Transformers, Prompt Engineering
- Model Fine-tuning & Evaluation
- Embeddings, RAG Pipelines, Semantic Search

Machine Learning & Deep Learning

- Model Training & Validation, Feature Engineering, CNNs, RNNs, LSTMs

Programming & Frameworks

- Python, FastAPI, LangChain, Hugging Face, TensorFlow, Scikit-learn

MLOps & Deployment

- Docker, REST APIs, AWS (EC2, S3, Lambda), AutoScaling, HomeLabbing

Tools & Platforms

- FAISS, Chroma, LlamaIndex

Professional Experience

AI Engineer

Rupadi International Pvt Ltd | Pune

Mar 2025 – Present

- Designed and deployed an **LLM-powered AI interviewer system** using LangChain, Redis, and Python for adaptive conversational assessments.
- Built **Transformer-driven pipelines** for real-time question generation and semantic analysis.
- Developed **modular FastAPI microservices** to support scalable inference and seamless backend integration.
- Collaborated cross-functionally with backend and frontend teams to deliver production-ready AI solutions.

Junior Data Scientist

Teamlease | Viman Nagar, Pune

Aug 2024 – Dec 2024

- Implemented **Retrieval-Augmented Generation (RAG)** pipelines using Hugging Face Transformer models.
- Designed robust **text preprocessing workflows** for large-scale unstructured datasets.
- Conducted **fine-tuning and evaluation** of generative models using BLEU, ROUGE, and accuracy metrics.
- Supported containerized ML workflows with Docker for reproducibility and experimentation.

Data Researcher

Collegedunia | Pune

Mar 2023 – May 2024

- Automated data extraction pipelines using **BeautifulSoup and Selenium**.
- Contributed to a **document-based generative chatbot** using FAISS and RAG techniques.
- Ensured data quality through validation, labeling, and source verification.

Key Projects

AI Interviewer Application

- Built a **multimodal AI interviewer chatbot** integrating LLMs, speech-to-text, and FastAPI microservices.
- Deployed on AWS with Redis caching for improved performance and scalability.

Financial Insights Bot

- Developed an **LLM-based financial analysis assistant** with a RAG layer for accurate, context-aware responses.

Admission Query Chatbot

- Implemented a **Transformer-powered educational chatbot** using FAISS-based semantic retrieval.

Education

- **B.E. Electrical Engineering** – SB Patil College, Indapur (2017–2020)
- **Diploma in Electrical Engineering** – KBP Polytechnic, Satara (2014–2017)

Hobbies & Interests

- **Building Home Labs** for experimenting with self-hosted services, networking, and ML deployments
- **Learning New Technologies & Skills**, especially in AI, cloud, and system design
- **Solving Complex Problems**, with a strong curiosity for optimization and efficiency